

Stage 1: Discovery Workbook

Stage One: Discovery

Forward Deployed AI Engineering — Capstone Project

The workbook you complete before you design anything. Fill this in first. Nothing in the later stages is valid unless it can be traced back to something you recorded on these pages.

Week One · Days One to Three

How to Use This Workbook

This workbook is a set of tables for you to complete. Work through the sections in order. Section one establishes who you are talking to, sections two to five gather the evidence, and section six is where you turn that evidence into a problem statement. Do not attempt section six before the others are genuinely finished.

What finished looks like: *You are finished with discovery when a further interview tells you nothing you did not already know, and when every claim in your problem statement can be pointed at a specific row in one of these tables.*

Discovery repeats until the answers stop changing.

Section 1: What the Interviews Tell You

Your discovery evidence is in two places: the five stakeholder interview transcripts in the datasets folder, and the ticket data itself. Start with the transcripts. Read all five before you fill in the table below.

For each person, record what you were trying to learn from them, what they actually told you, and what they seemed not to know about their own operation. That last column is often where the useful material is, because each of the five can only see their own part of the problem.

Who	What they told you	What they appear not to know	What you want to verify in the data
Marcus Adeyemi, Head of Support	FCR is 42% (dev data: 43.8%), industry benchmark is 65%. Response time is 8–12 hrs against a 2-hr SLA. 500+ tickets per week, 6 agents. Canned response library failed — finding the right canned response took longer than writing an answer from scratch. Primary fear: automated system sends something wrong. Secondary concern: enterprise customers notice degraded service. Autumn compliance review requires explainability.	Does not know his ticket distribution by intent or channel (“I would be guessing”). Unaware that <code>data_residency</code> and <code>data_export</code> are jointly the top intent (5.8% each) and that <code>security_incident</code> and <code>compliance_request</code> — both 100% must-not-auto-respond — sit in the top six. Does not know the <code>docs_comment</code> channel exists as a distinct volume source. His FCR figure (42%) differs slightly from the data record (43.8%).	Is 42% FCR accurate against <code>history.first_contact_resolution</code> ? What are the actual top intents? Which intents carry the must-not-auto-respond flag?
Sofia Restrepo,	Sorts queue by age (oldest = SLA breach risk). About 70% of tickets are ones she	Does not know that <code>docs_comment</code> is a distinct channel with the worst operational	Is 70% answerable from docs accurate? Do non-fluent tickets show

Who	What they told you	What they appear not to know	What you want to verify in the data
Tier One Agent	already knows the answer to. Search is too painful to use; all agents maintain personal answer files. Documentation quality is not the problem — findability is. Non-fluent English tickets take longer because she has to decode the question first; sometimes she answers the wrong question and the ticket goes round again. Prefers a draft-and-source tool over full automation — wants the draft and the relevant page, not an automated send.	metrics but the highest CSAT. Does not know that the non-fluent CSAT differential she describes (worst scores) is not visible in the development data — non-fluent tickets score marginally higher (3.04 vs 2.95). The handling-time cost she describes is real (+14%); the CSAT penalty is not.	lower CSAT? What is the actual resolution time differential by language fluency?
Daniel Okonkwo, Tier Two Engineer	About half of escalations could have been resolved at tier one given confidence or a findable page. Escalations arrive with no context — just the forwarded original thread; his first task is always to re-read everything. Personal snippet files are a risk: answers that were correct two years ago are still circulating. Security incidents, billing disputes, and data location must never be automated. Would work twice as fast if escalations arrived with a structured context packet.	Does not know that <code>docs_comment</code> tickets escalate at 65.4% — the highest rate of any channel. Does not know that <code>feature_request</code> and <code>unclear_request</code> are also 100% must-not-auto-respond (7% of volume combined). Does not know that 92 tickets in the development set were labelled <code>auto_respond</code> but actually escalated — the confidence and findability failures he describes quantified.	Does the data confirm his “about half” estimate for answerable-but-escalated tickets? Use <code>history.escalated = TRUE</code> × <code>labels.answerable_from_docs = TRUE</code> .
Ines Varga, Technical Writer	29 reviewed articles in the support knowledge base. Agents reconstruct answers from memory rather than copying her articles — she can tell by the near-verbatim phrasing. Internal search is keyword/title-based; customers describe problems in plain language. Wants source attribution on every answer so she can distinguish article errors from retrieval errors. Has asked for agent personal files to be consolidated and reviewed twice without result.	Does not know that <code>docs_comment</code> tickets — customer comments on her own articles — are a distinct support channel generating 78 tickets (15.6% of development volume) with the worst FCR and highest escalation rate of any channel. Does not know that 71.4% of development tickets are already labelled answerable from docs — coverage is not the gap. Does not know that <code>data_residency</code> (top intent by volume) has a 69% escalation rate and average resolution of 515 minutes.	Do answerable tickets have elevated repeat contact rates, indicating customers cannot locate answers that exist?
Ravi Menon, Customer	Service is slow but good when it arrives. Urgency is not uniform: a how-to question can wait; a deployment failure at 9am cannot. Searches the documentation himself; finds the answer about half the time. When the reply arrives it says the same thing and he has burned two hours. Wants transparency: knowing it is automated means he will verify before acting. Believes enterprise tier gets answers “in about an hour.”	His belief that enterprise gets faster service is contradicted by the development data: enterprise has the longest resolution time (484 min), lowest FCR (37.3%), and highest escalation rate (62.7%). His own business tier resolves fastest (359 min, 48.2% FCR). The regional performance gap — Latin America and to a lesser extent Asia-Pacific scoring worst — is invisible to him.	Does <code>customer_tier = 'enterprise'</code> show faster resolution than <code>'business'</code> ? What does the regional breakdown of CSAT and resolution time show?

Where the Accounts Disagree

At least three points of disagreement run through the transcripts, and each can be settled by going to the ticket data rather than by deciding whose account sounds more convincing.

The disagreement	Who says what	How the data settles it	What follows from the answer
What is inside escalations —	Marcus: escalation rate is a volume and capacity	<code>history.escalated = TRUE</code> × <code>labels.answerable_from_docs</code> :	The escalation rate is artificially inflated. Reducing it does not require more headcount —

The disagreement	Who says what	How the data settles it	What follows from the answer
capacity problem or confidence/findability problem	problem. Daniel: about half of escalations could have been resolved at tier one with confidence or a findable page.	138/281 escalated tickets (49.1%) were answerable from docs. A further 92 labelled <code>auto_respond</code> were actually escalated. Daniel is correct.	it requires confident, fast retrieval at tier one. The system should reduce escalations, not just response time.
Marcus's stated priority (FCR) vs. the team's actual behaviour (response time)	Marcus: FCR is his real priority. Sofia: she sorts by age to avoid SLA breach; does not mention FCR.	All FCR tickets resolve in under 212 minutes; no ticket taking longer than 216 minutes achieves FCR. Speed and FCR are the same population. But 24.8% of Q2 (fast) tickets still fail FCR — fast response $\neq$ resolution.	The incentive structure (reporting response time) creates a hidden FCR gap. A system that drafts and routes correctly improves both simultaneously, but FCR must be tracked from <code>history.first_contact_resolution</code> directly — not inferred from response time.
Non-fluent ticket quality — CSAT claim	Sofia: non-fluent tickets have the worst satisfaction scores. Dev data: non-fluent CSAT is 3.04 vs fluent 2.95 — a marginal reversal.	The development dataset does not support Sofia's CSAT claim. Non-fluent tickets score slightly higher. The real differential is resolution time: 466 min vs 408 min (+14%).	Intervention should target handling time (better intent extraction, clearer question decoding), not CSAT-focused routing. Sofia's instinct that something is wrong is right; her diagnosis of where it shows up (CSAT) is not supported by the data.
Proportion of escalations answerable from documentation	Daniel: about half reach him unnecessarily. Marcus: does not describe the composition of escalations.	49.1% of actually-escalated development tickets have <code>answerable_from_docs = TRUE</code> . Daniel's "about half" is confirmed.	Routing improvement is as important as response generation. A system that retrieves confidently for the 49.1% will reduce tier-two load without any change to escalation policy.
Ravi's belief that enterprise tier gets faster service	Ravi: enterprise colleague gets answers "in about an hour."	Enterprise: 484 min avg, 37.3% FCR, 62.7% escalation — worst operational metrics. Business tier (Ravi's own): 359 min, 48.2% FCR. Enterprise CSAT is highest (3.05), reflecting ticket complexity not service speed.	The tier gap Ravi worries about widening is already inverted. The group most in need is standard tier — 253 tickets (51% of dev set), 43.1% FCR, 56.9% escalation.

What Nobody Said

Real stakeholders leave out the things so familiar to them that they no longer notice them.

What was never mentioned	Why you would have expected it	How you will check it
The <code>docs_comment</code> channel (15.6% of dev volume) has the worst operational metrics — 515 min avg, 34.6% FCR, 65.4% escalation — but the highest CSAT of any channel (3.17)	Ines owns the documentation and would be expected to know that customer comments on her articles generate support tickets. Marcus tracks channels as volume. Sofia and Daniel work through a unified queue with no channel breakdown.	Group tickets by channel and compare resolution time, FCR, escalation rate, and CSAT. The pattern — patients customers who wait longest and escalate most but are most satisfied when answered — identifies them as the highest-value target for auto-response. Already verified.
17.4% of dev tickets (87/500) carry <code>must_not_auto_respond = TRUE</code> — and four intent types are blocked at 100%	Daniel named three categories to protect. Marcus described his fear. Neither stated a number or identified that <code>feature_request</code> and <code>unclear_request</code> are also entirely off-limits (35 tickets, 7% of dev volume).	Count tickets with the must-not-auto flag, then break down by intent. The result defines the hard floor on automation scope: the system must classify intent before deciding whether to respond at all. Already verified: four intents at 100%.
Latin America is a clear performance outlier: CSAT	Region affects when customers submit tickets relative to the support team's working hours, which affects how long	Break CSAT, resolution time, and FCR by region, then cross-check against language fluency to see whether the regional gap disappears when language is controlled for. Already verified: Latin America's fluent customers (CSAT

What was never mentioned	Why you would have expected it	How you will check it
2.73, FCR 38.3%, escalation 61.7%	tickets sit unattended. Ravi is the only customer voice and does not identify his region.	2.64) score lower than its non-fluent customers (CSAT 3.00) — the gap is not explained by language. Asia-Pacific (CSAT 2.94) is close to Europe (3.01) in the dev data and is not a strong secondary outlier.
Tickets arriving at 07:00–08:00 UTC take the longest to resolve (573–591 min) — a queue-start performance cliff	Sofia mentions Mondays are the busiest day but nobody described what happens to overnight tickets when the team logs on.	Group tickets by the hour they were received and compare average resolution time across the day. If 07:00–08:00 is the worst window, the most likely cause is overnight tickets accumulating and entering service simultaneously at shift start. Already verified.

Section 2: What the Ticket Data Shows

Before you interpret anything, count it. Load `development_tickets.json` and fill in the table below from the data itself rather than from impression. Where a figure surprises you, write the surprise down; those are usually the most productive threads to pull.

What to measure	Your figure	Where the figure came from	What surprised you about it
Total tickets in the sample	500 development tickets. (80 additional validation tickets are excluded from discovery — held for evaluation.)	<code>SELECT COUNT(*) FROM tickets WHERE dataset='development'</code>	The 500/80 split is intentional. Every figure in this workbook uses development only.
Split by channel	email 42.4% (212), chat 31.0% (155), docs_comment 15.6% (78), forum 11.0% (55)	Channel breakdown filtered to development dataset	<code>docs_comment</code> is a distinct channel nobody named in the interviews. It has the worst FCR (34.6%) and longest resolution time (515 min) but the highest CSAT (3.17) — customers are patient and satisfied once answered, but rarely answered on first contact.
Split by intent category	22 distinct intents. Top: <code>data_residency</code> and <code>data_export</code> tied at 5.8% (29 each), <code>rollback_request</code> 5.6% (28), <code>deployment_failure</code> 5.4% (27), <code>security_incident</code> and <code>compliance_request</code> 5.2% each (26).	Intent distribution on development labels	Marcus named authentication, deployment, billing, and API. The actual joint-top intents are <code>data_residency</code> and <code>data_export</code> — neither was mentioned. Both <code>security_incident</code> and <code>compliance_request</code> are in the top six and are 100% must-not-auto-respond.
Split by urgency	medium 45.2% (226), high 29.2% (146), low 25.6% (128)	Urgency distribution on development labels	High urgency is less than a third of volume. A blanket urgency-first routing policy would still leave 45% of tickets — the medium group — competing without differentiation. Intent matters more than urgency for routing decisions.
Proportion marked resolved on first contact	43.8% (219/500)	<code>history.first_contact_resolution</code> on development tickets	Marcus cited 42%. The data shows 43.8% — a 1.8pp discrepancy. The gap to the 65% industry benchmark is ~21 percentage points. If his reporting is not drawn directly from this field, system improvements may not register in what he presents to the executive team.
Average satisfaction rating	2.97 / 5	<code>AVG(history.csat_rating)</code> on development tickets	FCR tickets average 3.41; non-FCR tickets average 2.63 — a 0.78pp gap. The aggregate CSAT Marcus reports is a blend of two very different populations. Improving FCR is the

What to measure	Your figure	Where the figure came from	What surprised you about it
			highest-leverage path to improving the number he presents upward.
Most frequent single question	<code>data_residency</code> and <code>data_export</code> tied at 29 tickets each (5.8%)	Intent counts on development labels	Neither intent was named by Marcus. <code>data_residency</code> carries a 69% escalation rate and 515 min avg resolution — the highest-effort answerable category. It is the top-volume intent and the one most in need of reliable retrieval.
Proportion answerable from existing documentation	71.4% (357/500)	<code>labels.answerable_from_docs</code> on development tickets	Sofia's "seven in ten" estimate is accurate to within rounding. The answers exist. The gap between 71.4% and Ravi's ~50% self-service rate is 21 percentage points of pure findability failure — not a documentation coverage problem.
Proportion marked non-fluent, and their outcomes	24.0% of dev tickets (120/500). Non-fluent CSAT: 3.04. Fluent CSAT: 2.95. Resolution time: 466 min (non-fluent) vs 408 min (fluent), +14%. Escalation: 54.2% vs 56.8%. FCR: 45.8% vs 43.2%.	Language fluency × history metrics on development tickets	Sofia said non-fluent tickets have the "worst satisfaction scores." The dev data shows the opposite: non-fluent tickets score marginally higher. The real differential is resolution time (+14%). Sofia's instinct that something costs more is correct; her identification of CSAT as the signal is not supported.
Outcomes by customer tier	business: 359 min, 48.2% FCR, CSAT 2.91 (n=164). standard: 442 min, 43.1% FCR, CSAT 2.98 (n=253). enterprise: 484 min, 37.3% FCR, CSAT 3.05 (n=83).	<code>customer_tier</code> × history metrics on development tickets	Enterprise has the worst operational metrics but the best CSAT. Ravi's assumption (enterprise = faster service) is wrong. Standard tier is the largest and worst-served group on operational metrics — the primary improvement target.
Proportion that are repeat contacts	21.6% (108/500)	<code>history.repeat_contact</code> on development tickets	Answerable tickets: 16.2% repeat contact rate. Non-answerable: 35.0%. 16% of tickets whose answers exist in the docs still generate a follow-up — the first response is not pointing customers to the source clearly enough.

Two rows matter more than the rest: The proportion of tickets already answerable from existing documentation tells you whether CloudServe lack answers or merely lack a way of reaching them. Those are different problems with different solutions. The non-fluent segment matters for a different reason: something in the transcripts points at it, and the data will tell you whether that person was right.

Section 3: Where the Time Actually Goes

Understanding volume is not the same as understanding effort. A category that accounts for five per cent of tickets but forty per cent of agent hours is where the real problem lives.

Ticket category	Share of volume	Estimated share of effort	Why the two differ	Evidence for your estimate
<code>security_incident</code>	5.2%	~12–13%	100% escalated, avg 786 min — nearly double the 422 min dataset average. Every ticket requires tier-two handling; none can be auto-responded.	Dev avg resolution: 785.9 min. Escalation rate: 100%.

Ticket category	Share of volume	Estimated share of effort	Why the two differ	Evidence for your estimate
compliance_request	5.2%	~11–12%	100% escalated, avg 679 min. Compliance tickets require human judgment and potentially legal sign-off before any response is sent.	Dev avg resolution: 679.3 min. Escalation rate: 100%.
data_residency	5.8%	~10–11%	Not must-not-auto, but 69% escalation and 515 min avg. Agents cannot answer confidently without knowing the customer's specific contract and region.	Dev avg resolution: 515.0 min. Escalation rate: 69.0%.
unclear_request	3.0%	~5–6%	100% escalated, 516 min avg. The ticket cannot be processed until intent is established — every one requires a clarification round-trip before resolution begins.	Dev avg resolution: 515.7 min. Escalation rate: 100%.
High-frequency answerable (billing, rollback, API, account)	~24% combined	~15%	These are the tickets Sofia resolves in 4–5 minutes from memory. Low per-ticket effort, high volume — the category where retrieval automation delivers the highest ROI per ticket.	billing_query , rollback_request , api_usage_question , account_access all resolve under 420 min avg with lower escalation rates than the dataset average.

The Steps an Agent Takes on a Typical Ticket

Sofia describes her working process in some detail in the second transcript. Reconstruct it as a sequence of steps and record roughly where the time goes at each one. The steps that sound trivial are frequently where the minutes accumulate.

Step	What the agent does	Roughly how long	Could this be automated? Say why or why not
1	Open queue, sort by age to find tickets close to SLA breach	1–2 min per session	Partially. Urgency is already labelled. A system could surface breach-risk tickets automatically. Sofia's sort-by-age heuristic does not account for intent: a deployment_failure two hours old may matter more than a billing_query six hours old.
2	Read the ticket and decode the actual question — harder for non-fluent English	3–15 min depending on clarity	Yes, with caveats. An LLM can extract intent and restate the question clearly. For non-fluent tickets this step is where the +14% resolution time penalty accumulates. Intent extraction is the highest-value automation target for non-fluent tickets specifically.
3	Search for the answer — recall from memory, personal file, or (rarely) docs	2–10 min	Yes. This is the core retrieval problem. Sofia explicitly says she does not use official documentation because keyword search is too painful. Semantic retrieval replaces this step with a cited draft.
4	Compose and write the response	3–10 min	Yes, with a human review gate. Sofia's preferred workflow is a draft with the relevant page attached, not full automation. The draft is automatable; the send decision should remain with the agent for non-trivially risky responses.
5	Decide whether to escalate — based on confidence, not content	1–3 min	Partially. The system can flag low-confidence retrievals as escalation candidates. Currently 49.1% of escalations were answerable from docs — the decision is made on confidence, not on whether an answer exists. Surfacing the relevant article converts a confidence guess into a judgement call with evidence.
6	Forward ticket to tier two — no summary, no notes on what was tried	1 min	The forwarding is trivial; the missing context is not. Daniel: "If an escalation arrived saying here is the ticket, here is what I think it is about, here is the documentation that seemed relevant, and here is specifically what I was not confident about, I would be twice as fast." An automated context packet on every escalation is deliverable independently of auto-response.

Section 4: What the Client Counts as Success

You cannot improve a number nobody is watching. Marcus names the figures he reports and is explicit about which one he actually cares about, and they are not the same.

Measure	Who watches it	Current value	What they would call success	How confident are you in this?
First response time	Marcus (reports to exec team), Sofia (optimises her daily workflow around it)	8–12 hours average against a 2-hour SLA target	Under 2 hours consistently	High — Marcus stated it explicitly. The data contains full resolution time, not first-response time; the 8–12 hr figure cannot be independently verified from this dataset.
First contact resolution (FCR)	Marcus (calls it his “real priority”)	43.8% (dev data); Marcus stated 42%	65% (industry benchmark Marcus quotes)	High on target; medium on current value — Marcus’s stated 42% is a 1.8pp undercount against the data record.
CSAT rating	Marcus (reports to exec team — “red number for two quarters”)	2.97 / 5 (dev data)	customer mentions 4 but doesn’t look realistic from the dataset provided	High — from <code>AVG(history.csat_rating)</code> on dev set. The 2.97 figure is just below the implied target.
Escalation rate	Marcus (tracks as headline metric)	56.2% (281/500 dev tickets escalated)	Not stated; implied lower is better	High — <code>history.escalated</code> is the authoritative source. Marcus sees the rate but not what is inside the escalations.
Cost per ticket	Marcus (implicit — “every ticket that bounces to tier two costs roughly four times what a resolved one costs”)	Not in the data	Lower cost through reduced escalations and auto-draft responses	Medium — the 4x cost ratio is Marcus’s estimate, not a data-backed figure.
Accuracy / no wrong answers	Marcus (primary stated fear), Ravi (will screenshot a wrong answer)	Not measurable from current data	Zero confidently wrong automated responses	Low confidence in measurement — <code>must_not_auto_respond</code> is a design constraint, not a recall metric. <code>ground_truth_responses.json</code> enables evaluation on the answerable subset.

Project Brief Targets: Achievability Against Development Data

The project brief (§7) states baselines drawn from management estimates. Three of the five measures differ from the development data record. All achievability assessments below use the development data actuals as the true starting point.

Target (Brief §7)	Brief Baseline	Dev Data Actual	Brief Target	Achievable?	Evidence and ceiling
FCR ≥ 60%	42%	43.8%	60% or better	Yes — ceiling ~71%	49.1% of escalations (138/281 tickets) are answerable from docs. Converting those gives FCR = (219+138)/500 = 71.4%. The 60% target is a conservative waypoint well below the realistic arithmetic ceiling.
First reply < 5 min	8–12 hr	Not in data	Under 5 minutes	Yes — for 82.6% of volume	First-reply timestamp is absent from the dataset; only full resolution time is recorded. For the 82.6% of tickets without <code>must_not_auto_respond</code> , an auto-response pipeline can deliver first reply in seconds. The 17.4% that route to humans have unverifiable first-reply times from this data alone.
CSAT ≥ 4.0	3.2 / 5	2.97 / 5	4.0 or better	No — mathematical	FCR tickets average 3.41 CSAT; non-FCR tickets average 2.63. At 100% FCR the blended average is 3.41 — 0.59 below the target. The brief baseline (3.2)

Target (Brief §7)	Brief Baseline	Dev Data Actual	Brief Target	Achievable?	Evidence and ceiling
				ceiling is 3.41	also overstates the dev data starting point (2.97) by 0.23 points, meaning the required lift is 1.03 points, not 0.8. No routing or retrieval improvement can close this gap; the underlying reasons FCR tickets score below 4.0 are not addressed by faster or more accurate routing.
Escalation rate ≤ 30%	58%	56.2%	30% or lower	Yes — floor ~28.6%	Removing the 138 answerable escalations gives escalation = $143/500 = 28.6\%$, just inside the target. This is the arithmetic floor assuming perfect recall on answerable tickets; actual system performance approaches this ceiling as confidence calibration matures.
Repeat contacts halved	Not measured	21.6% (108/500)	≤ 10.8%	Yes — modelled floor ~10%	Answerable tickets with properly cited responses reduce repeat contact to near the non-answerable baseline. Modelled floor based on citation-quality responses: ~10.0%, just inside the 10.8% target.

Governance Conditions: Pre-Build Status

The governance conditions in Brief §7 are binary — they either hold or they do not. Two conditions cannot yet be evaluated before a system is built. One is already violated in the baseline data.

Condition	Requirement	Pre-Build Status	Evidence
Private data in outbound responses	Zero occurrences	Not yet evaluable — no system exists	PII present in <code>customer_name</code> , <code>customer_id</code> , <code>customer_region</code> fields. Prevention requires: retrieve from <code>documents</code> table only, never from ticket history; audit prompts for PII injection vectors.
Quality across customer groups	< 5pp difference between groups	Already violated at baseline — 10.5pp gap	North America FCR: 48.8%. Latin America FCR: 38.3%. Gap: 10.5pp — more than double the 5pp threshold before any system is built. A system that simply preserves the existing gap is not compliant. Closing the gap, not just not widening it, is an explicit design requirement.
Decision logging	Complete coverage	Not yet evaluable	Every routing decision must be reconstructable months later. Schema design must capture input, prediction, confidence, sources, action, and reason from day one.
Confidence calibration	Within 5pp of observed accuracy	Not yet evaluable	Classifier calibration requires evaluation data. The threshold is in scope; measurement begins once the classifier exists.

The Sentence Test

Complete the following sentence using only figures you recorded above. If you cannot complete it, your discovery is not finished.

Prompt	Your answer
This project will have been worth doing if, within three months of launch, ...	FCR has risen from 43.8% toward the 65% benchmark; first-response time has fallen below the 2-hour SLA on at least 80% of tickets; and the escalation rate for tickets labelled <code>answerable_from_docs = true</code> has fallen materially — meaning the 49.1% of escalations that were answerable from documentation are being resolved at tier one.
We will know it did not work if we see ...	The escalation rate holds steady or rises; CSAT stays below 3.0; or any measurable rate of auto-responded tickets where the customer's follow-up indicates the answer was wrong or irrelevant. The second failure Marcus described — more work for agents than it saves — would show as increased average handling time per ticket after launch.
The measure the client will actually be judged on internally is ...	First response time against the 2-hour SLA, because it is “what is in the agreement so it is what gets reported” (Marcus). FCR is Marcus's stated real priority but is not in the SLA and is not what the executive team currently sees. Any success story must show improvement on the reported metric, not only the preferred one.

Section 5: Data, Constraints and Risk

Everything you build depends on data that already exists, and every piece of that data carries constraints.

Data source	What it contains	Quality and gaps	Access constraints	Private data present?
Support tickets (<code>tickets</code> , <code>labels</code> , <code>history</code>)	500 development tickets with channel, intent, urgency, customer profile, resolution outcomes, and explicit <code>must_not_auto_respond</code> flags	High quality for labelled fields. No first-response timestamp — only full resolution time. 15 tickets (3.0%) are <code>unclear_request</code> , meaning intent was indeterminate even for the human labeler.	Supabase PostgreSQL; credentials in <code>.env</code>	Yes — <code>customer_name</code> , <code>customer_id</code> , <code>customer_region</code> , <code>customer_tier</code> . PII must not appear in automated responses or logs.
Documentation (<code>documents</code> , <code>document_relations</code>)	29 reviewed knowledge base articles across 7 categories. Freshness tracked via <code>last_reviewed_days_ago</code> .	Ines confirms the 29 articles are accurate within their review cycle. Gaps: feature requests, roadmap questions, and novel incidents have no retrievable answer. <code>data_residency</code> — the top-volume intent — carries a 69% escalation rate, suggesting the articles in this category may not fully resolve the questions customers are actually asking.	Same database	No PII — product documentation only.
Past resolutions (<code>ground_truth_responses.json</code>)	Reference responses for a subset of development tickets, with <code>must_mention</code> and <code>must_not_claim</code> constraints. Written by <code>senior_support_agent</code> .	Suitable for evaluation. Does not cover the full ticket set. Does not include any <code>must_not_auto_respond</code> tickets by definition.	Flat file in <code>requirements/05_Datasets/</code>	No PII
Agent personal files	Answer snippets accumulated by Sofia, Daniel, and others over months to years	Not in any system. Unreviewed. Daniel: "There are answers circulating in people's personal snippet files that were correct two years ago and have not been right since."	Not accessible — not digitised or centralised	Unknown — agent-authored, no governance
Customer records	Customer tier, region, language fluency in the ticket table	No contract details, no SLA tier specifics, no account history beyond what is in individual tickets	Same database	Yes — <code>customer_tier</code> and <code>customer_region</code> may be contractually sensitive

Initial Risk Register

List what could go wrong once this system is speaking to customers without supervision. You will develop these further in stage five, but the first pass belongs here.

What could go wrong	How likely	How bad	Who it affects	First thought on preventing it
The system answers confidently and incorrectly	Medium — retrieval can surface a relevant-seeming but wrong article, especially for <code>data_residency</code> (69% escalation, 515 min avg — agents get this wrong even without automation) and compliance topics	Very high — Marcus: “They will screenshot a confidently incorrect answer and put it on the internet within the hour.” Ravi: “I will act on it and break something.”	All customers; enterprise customers most exposed given ticket complexity	Constrain auto-response to tickets where <code>must_not_auto_respond = FALSE</code> , retrieval confidence is above threshold, and the retrieved article was reviewed recently. Show source citation on every response.
Private information appears in a reply	Low — tickets contain <code>customer_name</code> , <code>customer_id</code> ; retrieval is from documentation not tickets	High — GDPR/contractual risk; Marcus flagged a compliance review in autumn	Any customer whose data appears	Never retrieve from ticket history. Retrieve only from the curated <code>documents</code> table. Audit the prompt for PII injection vectors.
Some customers get consistently worse answers	Medium — Latin American (CSAT 2.73, FCR 38.3%) customers already receive the worst service; automation that does not account for timezone and queue-start patterns will replicate this	High — systematic quality gap for a minority is a reputational and fairness risk	Latin American customers primarily; also <code>docs_comment</code> channel users if not prioritised	Evaluate system performance segmented by <code>language_fluency</code> , <code>customer_region</code> , and <code>channel</code> — not just overall accuracy. An aggregate pass rate can hide a segment that consistently fails.
The documentation the system relies on goes out of date	Medium — 29 articles on rotation; <code>last_reviewed_days_ago</code> is tracked; novel incidents and product changes are not covered until Ines writes a new article	High for compliance-adjacent topics — wrong information in these areas is a regulatory risk	All customers; compliance-sensitive customers most exposed	Block auto-response for articles above a staleness threshold. Flag stale articles to Ines automatically. Build a feedback path from agent corrections back to the knowledge base — the current channel failed (Conflict 4).
Personal snippet file content enters the retrieval corpus	High if snippet files are consolidated without review — Ines has asked for this twice without result	High — Daniel: “If you learn from those you will scale up a mistake”	All customers	Retrieve only from the reviewed <code>documents</code> table. Do not ingest agent personal files without Ines’s review and approval. Treat consolidation as a separate, governed editorial project.
The system responds to <code>must_not_auto_respond</code> tickets	Low if intent classification is reliable; medium if not (15 <code>unclear_request</code> tickets had indeterminate intent even for human labelers)	Very high — security, compliance, and billing disputes must not receive automated commitments	Enterprise customers; legally exposed customers	Intent classification must be the first step, before retrieval or response. Must-not-auto intents route to a human queue with no automated draft. For <code>unclear_request</code> , the system should ask for clarification rather than attempt retrieval.

Section 6: The Problem Statement

Now, and only now, write down what the problem actually is. Every clause of what you write must be supported by a row you have already filled in, and you are asked to cite those rows explicitly. This is the paragraph your entire project rests on.

Element	Your statement	Which section and row supports it
What the client asked for	An automated system that answers easy tickets so agents can handle hard ones, primarily to reduce first-response time below the 2-hour SLA.	Section 4 — Marcus: "I picture it answering the easy ones so my people can do the hard ones... I need for the response time number to come down."
What the evidence suggests they actually need	A retrieval-assisted triage system that classifies intent first, surfaces the relevant documentation with a citation, and produces a reviewed draft for agent approval — routing the 17.4% of must-not-auto tickets immediately to humans and attaching a structured context packet to every escalation.	Section 2 (answerable_from_docs 71.4%, must_not_auto 17.4%); Section 1 — Sofia: "If it hands me a ticket with a draft and the relevant page attached, that would genuinely help"; Section 3 Step 6 — Daniel's context packet request.
The gap between those two	Full automation is unsafe for 17.4% of tickets and fragile for others where retrieval confidence will be low. The real leverage is not answering automatically — it is getting the right article in front of the right agent in the first 30 seconds, which currently does not happen because internal search is keyword-based and agents have stopped using it.	Section 2 (must_not_auto 17.4%); Section 1 — Ines: "Someone writes 'my deployment keeps dying' and my article is called 'resolving container health check failures.' There is no path between those two phrases in a keyword search."
Who is affected and how	All 500 dev tickets are affected by slow first response. The 357 answerable-from-docs tickets (71.4%) are the primary addressable population. Non-fluent speakers (24%, 120 tickets) take 14% longer — intent extraction is the specific fix. Latin American customers receive the worst service across all metrics, likely due to timezone coverage. docs_comment users (15.6%) are the highest-value channel for auto-response.	Section 2 (non-fluent resolution time, channel split); Section 1 blind spots (regional gap, docs_comment).
What will change if this is solved	FCR rises toward 65% as the 49.1% of escalations that were answerable from docs are resolved at tier one. Repeat contact rate falls from 21.6% as responses include citations customers can self-serve from. Handling time falls for the 71% of tickets Sofia already knows the answer to — draft generation replaces the manual search step.	Section 2 (FCR 43.8%, repeat contact 21.6%, answerable 71.4%); Conflict 1 verdict (49.1% of escalations answerable); Section 3 Step 3.
What is explicitly not in scope	Feature requests (no answer to retrieve), novel incidents (no article exists yet), security_incident and compliance_request tickets (must have human sign-off regardless of retrieval quality), roadmap and timing questions, and the agent personal snippet file consolidation (a separate editorial project that must precede any corpus expansion).	Section 5 data constraints — Ines: "Feature requests, obviously, since there is no answer to retrieve"; must_not_auto_respond register; risk register row 5.

The One Paragraph Version

Write the problem in a single paragraph, in language the head of support would recognise and agree with. No technical vocabulary. If you cannot express it without mentioning retrieval or embeddings, you are describing your solution rather than their problem.

CloudServe's support team is answering the same questions repeatedly, slowly, without looking at the documentation that already contains the answers. Seven in ten tickets are ones the agents already know the answer to, yet average resolution takes over seven hours because finding and writing the answer is manual every time. Nearly half of the tickets that reach tier two — at four times the cost of a tier-one resolution — could have been answered at tier one if the agent had been able to find the right page quickly and confidently. Four of the five project brief targets are within reach from the development data: a retrieval-assisted system that correctly handles answerable tickets can bring FCR from 43.8% to above 60%, drive escalation from 56.2% to below 30%, cut repeat contacts from 21.6% to below 10.8%, and deliver auto-responded tickets a first reply in seconds rather than hours. The one target that cannot be reached through operational improvement alone is the 4.0 CSAT goal — FCR tickets themselves average only 3.41, so even perfect first-contact resolution produces a blended score of 3.41, not 4.0; the gap requires addressing satisfaction drivers that routing does not touch. There is also a problem that exists before any system is built: Latin American customers are already receiving service 10.5 percentage points worse than North American customers on FCR, more than double the five-point governance threshold the brief requires a deployed system to maintain. A system that only preserves that gap is not compliant; closing it must be a design requirement, not a tolerance to manage after launch.

Problem Statement

#	Statement
1	CloudServe's support team resolves only 43.8% of tickets on first contact, against an industry benchmark of 65%, because agents cannot find the relevant documentation quickly enough — not because the documentation is missing. 71.4% of tickets are already answerable from existing articles.
2	49.1% of tickets that reach tier two (at four times the cost of a tier-one resolution) were answerable from documentation. The escalation rate is inflated by a confidence and findability failure, not by genuine complexity.
3	Average resolution time exceeds seven hours against a two-hour SLA commitment. The delay is not in composing answers — it is in the manual search step that precedes composition, which agents have abandoned in favour of personal memory and unreviewed snippet files.
4	17.4% of tickets (87/500) must never receive an automated response — covering security incidents, compliance requests, data residency queries, feature requests, and unclear requests. Any automation that does not classify intent first is unsafe before it starts.
5	The CSAT target of 4.0 cannot be reached through operational improvement alone. FCR tickets average 3.41 CSAT; even at 100% first-contact resolution the blended score ceiling is 3.41. The 4.0 target requires addressing satisfaction drivers that faster or more accurate routing does not touch.
6	Latin American customers are already receiving service 10.5 percentage points worse than North American customers on FCR (38.3% vs 48.8%), exceeding the project brief's five-point governance threshold before any system exists. Closing this gap is a design requirement, not a post-launch tolerance.
7	The agent personal snippet files are an unreviewed corpus of answers — some years out of date — circulating outside any quality control. If those files enter the retrieval system without editorial review, the system will scale up errors that already exist.
8	The documentation channel (docs_comment) accounts for 15.6% of development volume, has the worst operational metrics (34.6% FCR, 65.4% escalation, 515 min avg), and the highest CSAT (3.17) — identifying its customers as the most patient and highest-value target for auto-response. No stakeholder named this channel unprompted.

Before you move to Stage Two: Read your problem statement back against each of the five transcripts in turn and ask whether that person would recognise it as a description of their situation. If any of the five would object, either your framing is off or you have found something they cannot see, and you should be able to say which. A statement that all five would simply agree with is usually one that has not gone far enough.

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